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United States Patent	12398533
Kind Code	B2
Date of Patent	August 26, 2025
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Continuous calibration of grade control system

Abstract

A work machine includes a frame, a linkage assembly, a work implement connected to the linkage assembly, and a grade control calibration system. The grade control calibration system includes a vision processing system, which includes a sensor fixed to the linkage assembly and a first controller. The vision processing system is configured to measure a length of a cutting portion of the work implement, and to transmit the length of the cutting portion of the work implement. The grade control calibration system also includes a grade control system in communication with the vision processing system and the linkage assembly. The grade control system includes a second controller configured to receive the length of the cutting portion of the work implement from the first controller, and to calibrate a position of the work implement based on the received length of the cutting portion of the work implement.

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Appl. No.:	17/725931
Filed:	April 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230340755 A1	Oct. 26, 2023

Publication Classification

Int. Cl.: E02F9/26 (20060101); E02F3/16 (20060101); E02F3/43 (20060101); E02F3/84 (20060101); E02F3/90 (20060101); G06T7/00 (20170101)

U.S. Cl.:

CPC **E02F3/907** (20130101); **E02F3/16** (20130101); **E02F3/842** (20130101); **E02F3/847** (20130101);

Field of Classification Search

CPC: E02F (3/907); E02F (3/16); E02F (3/842); E02F (3/847); E02F (3/844); E02F (3/7618);
E02F (3/7613); E02F (3/32); E02F (9/265); E02F (9/268); G06T (7/0006); G06Q (10/20)

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to a grade control system, and more particularly, to a method and system for continuous calibration of the grade control system of a work machine based.

BACKGROUND

(2) Conventional earthmoving machines such as track-type tractors, motor graders, scrapers, and/or backhoe loaders, may include a ground-engaging implement, such as a dozer blade or bucket, which may be used on a worksite in order to alter a geography or terrain of a section of earth. The implement may be controlled by an operator and/or by an autonomous grade control system. To achieve a final surface contour or a final grade, the implement may be adjusted to various positions by the operator or the grade control system. Accurately positioning the implement, however,

requires knowledge by the grade control system and/or operator of the machine as to the specific dimensions of the implement and its components.

(3) For example, the ground-engaging implements described above may include earth cutting portions such as teeth, shrouds, and/or lips. These earth cutting portions may wear more quickly than underlying equipment, as they initiate contact with the ground surface before the body of the excavating bucket and may encounter highly abrasive materials. These conditions cause the earth cutting portions to erode and, eventually, to wear out or fail. If not regularly updated, the grade control system will inaccurately position the implement during grading, resulting in an incorrect surface contour or grade. Manual calibration events are typically performed at certain intervals, for example, monthly, to measure various components of the implement, including the earth cutting portions, and to update the dimensions in the grade control system. However, manual calibration is tedious, time consuming, and costly for operators and maintenance personnel.

(4) Prior art attempts to determine wear associated with a part rely on determining when a part needs replacement. For example, U.S. Pat. No. 9,613,413 describes systems and methods for determining part wear using a mobile device. The system relies on capturing digital images using a camera on the mobile device and determining wear using a differential number of pixels between the imaged part and a simulated surface of an unworn part. A percentage or degree of wear is determined based on these distances. According to the degree of wear, an alert or warning is displayed to a user indicating replacement for the part is necessary.

(5) In light of the foregoing, a need exists for continuous calibration of grade control systems.

SUMMARY

(6) In accordance with one aspect of the present disclosure, a work machine is disclosed. The work machine may include a frame, a linkage assembly, a work implement connected to the linkage assembly, and a grade control calibration system. The grade control calibration system may include a vision processing system, which may include a sensor fixed to the linkage assembly and a first controller. The vision processing system may be configured to measure a length of a cutting portion of the work implement, and to transmit the length of the cutting portion of the work implement. The grade control calibration system may also include a grade control system in communication with the vision processing system and the linkage assembly. The grade control system may include a second controller configured to receive the length of the cutting portion of the work implement from the first controller, and to calibrate a position of the work implement based on the received length of the cutting portion of the work implement.

(7) In accordance with another aspect of the present disclosure, a grade control calibration system for a work machine is disclosed. The work machine may include a frame, a linkage assembly and a work implement. The grade control calibration system may include a vision processing system and a grade control system. The vision processing system may include an imaging device fixed to the linkage assembly and configured to generate a three-dimensional point cloud of a cutting portion of the work implement, and a vision controller in electronic communication with the imaging device and configured to calculate a current length of the cutting portion of the work implement based on the three-dimensional point cloud generated by the imaging device. The vision controller may transmit the current length of the cutting portion of the work implement. The grade control system may be in communication with the vision processing system and the linkage assembly and may include a grading controller configured to receive the current length of the cutting portion of the work implement from the vision controller and to calibrate a position of the work implement relative to the terrain based on the received length of the cutting portion of the work implement.

(8) In accordance with yet another aspect of the present disclosure, a method of grading terrain using a work implement of a work machine. The work machine may include a linkage assembly and a sensor coupled to the linkage assembly. The method may include detecting the work implement within a field of view of the sensor; identifying, by the sensor, a cutting portion of the work implement, calculating, by a controller electronically coupled to the sensor, a length of the

cutting portion of the work implement; calibrating, by the controller, a grade control system of the work machine based on the calculated length of the cutting portion of the work implement; maneuvering, by the calibrated grade control system, the cutting portion of the work implement proximate the terrain; and grading the terrain with the cutting portion of the work implement. (9) These and other aspect and features of the present disclosure will be better understood upon reading the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of an exemplary work machine, according to an embodiment of the present disclosure;
- (2) FIG. 2 is a schematic diagram of a grade control calibration system, according to an embodiment of the present disclosure;
- (3) FIG. 3 is a perspective view of a camera system of the exemplary work machine of FIG. 1, according to an embodiment of the present disclosure;
- (4) FIG. 4 is a front view of a work implement of the exemplary work machine of FIG. 1, according to an embodiment of the present disclosure;
- (5) FIG. 5 is a perspective view of a work implement of the exemplary work machine of FIG. 1, according to an embodiment of the present disclosure; and
- (6) FIG. 6 is a flowchart illustrating a method of grading, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(7) Reference will now be made in detail to specific embodiments or features, examples of which are illustrated in the accompanying drawings. Wherever possible, corresponding or similar reference numbers will be used throughout the drawings to refer to the same or corresponding parts.

(8) An exemplary embodiment of a work machine **10** is illustrated in FIG. 1. The work machine **10** may be, for example, an excavator, a wheeled or tracked loader, a mining shovel, a backhoe loader, or any other type of work machine known in the art. As an excavator, the work machine **10** may include a power source **12** (e.g., an engine, a motor, a battery bank, etc.) mounted to a frame **14** and configured to drive one or more ground-engaging elements **16** for propelling the work machine **10** across a worksite. The ground-engaging elements **16** may be mounted or movably coupled to the frame **14** and may include, for example, tracks, wheels, and/or combinations thereof. The frame **14** may also support an operator cab **18** configured to house, for example, an operator's seat and an operator console or other machine control devices for controlling various functions of the work machine **10**. In other embodiments, however, the work machine **10** may be an autonomous machine, a semi-autonomous machine, a remotely operated machine, or a remotely supervised machine, among others.

(9) The illustrated work machine **10** may also include a linkage system **20** and a work implement **22** connected to the linkage system **20**. The linkage system **20** may include a boom **24** pivotally connected to the frame **14**, a stick **26** pivotally connected to the boom **24**, and a linkage **28** pivotally connecting the work implement **22** and the stick **26**. The work implement **22** may include a bucket **30** pivotally connected to the linkage **28**. The bucket **30** may include a cutting portion **32** which, as illustrated, comprises a plurality of teeth **34**. In other embodiments, the cutting portion **32** may comprise a smooth edge.

(10) The work machine **10** may further include a sensor **40** configured to, among other things, measure a length of each of the plurality of teeth **34**. By way of example only, and not by way of

limitation, the sensor **40** may be an imaging device such as a smart camera or smart vision system, a monocular camera, an infrared camera, a high resolution camera, an array of one or more types of cameras, an opto-acoustic sensor, a radar, a laser based imaging sensor, or the like, or combinations thereof, configured to assist recognition, and monitoring of the work implement **22** and the worksite. The sensor **40** may be positioned on the work machine **10** to obtain a field of view **42** toward the work implement **22**. More specifically, the sensor **40** may be an imaging device positioned on the work machine **10** to capture images in their field of view **42** during operation of the work machine, and having a dedicated processor onboard, including video processing acceleration provided by a field-programmable gate array (FPGA), a digital signal processor (DSP), a general purpose graphics processing unit (GP-GPU), or any other suitable microprocessor with supporting application software, capable of determining depth and volume from real-time images or videos.

(11) As illustrated, the work machine **10** may include a left imaging device **40a** mounted on a left side **44** of the boom **24**, and a right imaging device **40b** (not shown) mounted on the right side (not shown) of the boom. In other embodiments, the work machine **10** may also include one or more imaging devices **40** mounted on a left side of the stick **26** and/or a right side of the stick. The imaging devices **40a**, **40b** may be stereo cameras. The left imaging device **40a** may continuously capture images and/or videos in its field of view **42a**, which may include worksite terrain primarily to the left side and in front of (in a direction of travel) the work machine **10** and at least a left half **46** (see also FIG. 4) of the work implement **22**. The right imaging device **40b** may continuously capture images and/or videos in its field of view **42b**, which may include worksite terrain primarily to the right side and in front of the work machine **10** and at least a right half **48** (see also FIG. 4) the work implement **22**.

(12) Referring now to FIG. 2, the work machine **10** may further include a grade control calibration system **100** including a vision processing system **200** and a grade control system **300**. The vision processing system **200** may include the left imaging device **40a**, the right imaging device **40b** and a vision controller **202**. The grade control system **300** may include a plurality of position sensors **302**, an inertial measurement unit (IMU) **304**, a global positioning system (GPS) unit **306** and a grade controller **308**. In one embodiment, the plurality of position sensors **302** are electrohydraulic position sensors associated with hydraulic components (e.g. cylinder or cylinder rods) of the boom **24**, the stick **26** and the work implement **22**, and configured to detect positional and/or velocity information associated with these components. Such linkage kinematics may be used to assist with calibrating a position of the work implement **22**, as will be discussed in further detail below.

(13) The grade controller **308** may be in electronic communication with the vision controller **202**, and both may communicate with a gateway controller **50**. The gateway controller **50** may, for example, provide a connection between the vision controller **202** and the grade controller **308** and at least one remote entity **52** via a network (not shown). In such an arrangement, the gateway controller **50** may transmit data to and receive data from the remote entity **52** via the network. The remote entity **52** may include a web server, computing device and/or storage device (e.g. database). In one embodiment, the remote entity **52** may be a database for storing a variety of information related to the grade control system **300** and the vision processing system **200**. More specifically, the database may store a list of machines, including the work machine **10**, each identified by a unique identifier. The table may associate each unique identifier with a set of data related to the components of each machine. For example, as to the work machine **10**, the table may associate a set of characteristics associated with the work implement **22**, such as a bucket **30** identifier or model number, an original number of teeth **34**, an original number of lip shrouds **64** (FIG. 4), an original length of each tooth, a 3D tooth profile, and a 3D shroud profile. This data may be accessed and updated by both the vision processing system **200** and the grade control system **300** through the gateway controller **50**.

(14) Each of the vision controller **202**, the grade controller **308** and the gateway controller **50** may

include any type of device or any type of component that may interpret and/or execute information and/or instructions stored within a memory (not shown) to perform one or more functions. The memory may include a random access memory (“RAM”), a read only memory (“ROM”), and/or another type of dynamic or static storage device (e.g., a flash, magnetic, or optical memory) that stores information and/or instructions for use by the example components, including the information and/or instructions used by the vision controller **202**, the grade controller **308** and the gateway controller **50** (as explained in further detail below). Additionally, or alternatively, the memory may include non-transitory computer-readable medium or memory, such as a disc drive, flash drive, optical memory, read-only memory (ROM), or the like. The memory may store the information and/or the instructions in one or more data structures, such as one or more databases, tables, lists, trees, etc. Finally, each of the vision controller **202**, the grade controller **308** and the gateway controller **50** may include a processor (e.g., a central processing unit, a graphics processing unit, an accelerated processing unit), a microprocessor, and/or any processing logic (e.g., an FPGA, an application-specific integrated circuit (ASIC), etc.), and/or any other hardware and/or software.

(15) Referring now to FIG. **3**, the left imaging device **40a** is illustrated. While only the left imaging device **40a** is illustrated in FIG. **3**, the description and discussion herein of the left imaging device should be understood to refer analogously to the right imaging device **40b**, except where otherwise indicated or apparent. As illustrated, the left imaging device **40a** may be a stereo camera module including a pair of monochrome lenses **53** and a color camera lens **54**. The monochrome camera lens **53** may capture, for each pixel of an image, an amount of light. In other words, the monochrome camera lens **53** may capture the image in black-and-white. Similarly, the color camera lens **54** may capture, for each pixel of an image, a color hue. In other words, the color camera lens **54** may capture the image in color. The left imaging device **40a** may also be capable of capturing images of its field of view in all levels of ambient light (i.e. both during the day and at night) with or without color.

(16) The left imaging device **40a** may be fixed to a mount **56**, which may be installed on a portion of the boom **24** of the work machine **10**. The mount **56** may be magnetic, and may thus be repositionable on the boom **24**, the stick **26**, the frame **14** or another area of the work machine **10** in order to adjust the field of view **42a** of the left imaging device **40a**. In other embodiments, the mount **56** may be a mount typically known in the art which integrates the left imaging device **40a** on the work machine **10**.

(17) The left imaging device **40a** utilizes a three-dimensional (3D) point cloud system, in which points within an image captured by the left imaging device (hereinafter referred to as a “stereo image”) are stitched together to generate a 3D cloud map from which the depth and/or distance of objects with respect to the environment and to each other may be determined. In operation, therefore, the left imaging device **40a** may capture a stereo image and generate a 3D cloud map of the worksite, the work implement **22**, and any other objects within the left imaging device's field of view **42a**. The vision controller **202** may use the stereo image to generate a disparity map, may overlay the 3D cloud map onto the disparity map, and consequently, measure relevant distances, depths, etc. More specifically, the vision processing system **200** may calculate a length of a tooth **34** of the work implement **22** by generating a 3D cloud map of the work implement **30**, identifying the specific region of the 3D cloud map that corresponds to the tooth **34** of the work implement, generating a disparity map, and overlaying the tooth region of the 3D cloud map on the disparity map to obtain a length value of the tooth.

(18) As shown in FIGS. **4** and **5**, and as noted above, the work implement **22** may comprise a bucket **30** with a cutting portion **32**. The bucket **30** may also generally include a plurality of heel shrouds **58**, a pair of side protectors **60**, and a pair of side cutters **62**. The cutting portion **32** may include a plurality of ground engaging tool assemblies or teeth **34** and a plurality of lip shrouds **64**. Each tooth **34** may include an adapter **66** configured to engage a base edge **68** of the cutting portion

32 of the bucket **30**. Each tooth **34** may also include a ground engaging tip **70** removably attached to the adapter **66**. The tip **70** may endure the majority of the impact and abrasion caused by engagement with work material, and may consequently wear down more quickly than, for example, the adapter **66**. As the grade control system **300** relies on knowing a length L of each tooth **34** to determine a position of the work implement **22** relative to the terrain of the work site, it is imperative that the grade control system utilizes an accurate length of each tooth in its calibration. Traditionally, machine operators, owners, servicemen, or dealers periodically observe the amount of erosion sustained or “wear level” of each tooth **34**, measure each tooth, and manually update the measurements associated with the grade control system **300**. However, this procedure can be costly, time consuming and inaccurate.

INDUSTRIAL APPLICABILITY

(19) The teachings of the present disclosure may find applicability in many industries including, but not limited to, earth moving equipment. In addition, the disclosed systems and methods may find application in any environment in which determining a length of a part is desired. The present solution reduces the time, energy and costs required to continually and manually measure a tooth of a work implement and calibrate a grade control system, and enables an accurate position of the work implement while grading. One skilled in the art will recognize, however, that the disclosed grade control calibration system could be utilized in relation to other machine components subject to wearing and erosion that may or may not be associated with a work implement or ground engaging tool.

(20) Referring now to FIG. **6**, and with continued reference to FIGS. **1-5**, a method **600** of grading using the present grade control calibration system is provided in flowchart format. Prior to operation of the work machine **10** at a worksite, a pre-existing three dimensional site map as well as a work tool profile may be retrieved by the work machine from the remote entity **52**. More specifically, an operator of the work machine **10** may use the operator console in the operator cab to select the specific model number of the bucket **30**. The profile associated with the bucket **30** may then be retrieved and loaded into the grade control system **300**. As noted above, the profile of the bucket **30** may include a bucket identifier or model number, an original number of teeth **34** installed on the bucket, an original number of lip shrouds **64** installed on the bucket, an original length of each tooth, a 3D tooth profile, and a 3D shroud profile.

(21) As the work machine **10** operates, the sensors **40** may continuously scan the work implement **22** and the environment around the work machine **10** as it moves about the worksite. To begin the calibration of the grade control system, and to properly measure the length L of the teeth **34** of the work implement **22**, the entire bucket **30** needs to be within the fields of view **42a**, **42b** of the left imaging device **40a** and the right imaging device **40b**. As illustrated in FIG. **5**, for example, the bucket **30** is within the fields of view **42a**, **42b** of the imaging devices **40**, with an area of overlap **72**. To achieve this alignment, the grade control system **300** and the vision processing system **200** monitor the linkage kinematics of the work machine **10**. Namely, data transmitted to the grade controller **308** from the plurality of position sensors **302**, as well as data provided by the IMU **304**, and global positioning data provided by the GPS unit **306** is monitored until the values of each linkage kinematic data enters a predetermined threshold (step **602**).

(22) Once the threshold values are met, the bucket **30** is considered aligned within the fields of view **42a**, **42b** of the left and right imaging devices **40a**, **40b**, and the imaging devices **40** may begin their detection process (step **604**). For example, the left imaging device **40a** would capture a stereo image of the bucket **30**, as illustrated in FIG. **5**, for example, and transmit the image to the vision processing controller **202** (step **606**). Simultaneously, the right imaging device **40b** would capture a stereo image of the bucket **30**, as illustrated in FIG. **5**, for example, and transmit the image to the vision processing controller **202** (step **606**). The vision controller **202** may use the stereo images to generate a depth or disparity map.

(23) Once received by the vision processing controller **202**, the images are analyzed and the

plurality of teeth **34**, the plurality of lip shrouds **64** and other implements (e.g. side cutters **62**) may be identified using machine learning algorithms, as generally understood in the art. The machine learning algorithms may be trained to detect positions of teeth **34**, lip shrouds **64**, and other implements and to generate a 3D point cloud outline of the teeth, shroud and other detected implements (step **606**).

(24) At a step **608**, the orientation of the teeth **34**, lip shrouds **64** and other implements are verified with the profile data associated with the model of the bucket **30**. For example, if the machine learning algorithms identify only two teeth **34**, but the bucket profile indicates this particular model of bucket **30** should have four teeth, then a discrepancy is noted. More specifically, at step **610**, to record any discrepancies, a non-detection counter is incremented and stored on the remote entity **52**. Once the non-detection counter exceeds a pre-determined threshold, an alert is generated and displayed to the operator of the work machine **10** indicating that the respective non-detected teeth **34**, lip shrouds **64** and/or other implement are missing.

(25) At a step **612**, the vision processing controller **202** may superimpose each 3D point cloud generated for detected teeth **34**, lip shrouds **64** and other implements over the previously generated disparity map. Then, with respect to each tooth **34** and lip shroud **64** specifically, the length L of each detected tooth and the length of each detected lip shroud is calculated from the 3D point clouds as measured inside the boundary outlined by the machine learning algorithms (step **613**). The measured length L of each tooth **34** is transmitted from the vision processing controller **202** to the grade controller **308**, which updates the current value of length L of each tooth **34** in the bucket **30** profile within the grade control system **300** (step **613**).

(26) Finally, the grade control system **300** utilizes the updated length L value of each tooth **34** in its configuration and positioning of the bucket **30** within the worksite. More specifically, the grade control system **300** utilizes the 3D site map and the bucket **30** profile data to accurately position the work implement **22**. With inaccurate tooth measurements, the grade control system **300** may believe a grade is deeper or shallower than its actual depth. This is important, for example, when water or gas pipes are present in the ground. As such, at step **614**, the grade control system **300** utilizes the updated current value of the length L of each tooth **34** to accurately position the work implement **22** within the terrain of the worksite, and according to defined boundaries of the 3D site plan.

(27) While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed machines, systems and assemblies without departing from the scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A work machine, comprising: a frame; a linkage assembly; a work implement connected to the linkage assembly; and a grade control calibration system including: a vision processing system including a sensor fixed to the linkage assembly and a first controller, the vision processing system configured to measure a length of a cutting portion of the work implement and to transmit the length of the cutting portion of the work implement, and a grade control system in communication with the vision processing system and the linkage assembly, the grade control system including a second controller configured to receive the length of the cutting portion of the work implement from the first controller and to calibrate a position of the work implement based on the received length of the cutting portion of the work implement.
2. The work machine of claim 1, wherein the linkage assembly includes a boom pivotally connected to the frame and a stick pivotally connected to the boom and the work implement.

3. The work machine of claim 1, wherein the work implement is a bucket including a plurality of teeth.
4. The work machine of claim 3, wherein the cutting portion of the work implement corresponds to the plurality of teeth and the sensor is configured to measure the length of each of the plurality of teeth.
5. The work machine of claim 3, wherein the work implement is a bucket including a smooth cutting edge, the smooth cutting edge extending a width of the bucket.
6. The work machine of claim 5, wherein the cutting portion of the work implement corresponds to the smooth cutting edge and the sensor is configured to measure the length of the smooth cutting edge along the width of the bucket.
7. The work machine of claim 1, wherein the sensor is a stereo camera that generates a three-dimensional point cloud of the cutting portion of the work implement.
8. The work machine of claim 7, wherein the stereo camera includes a monochrome lens and a color lens.
9. A grade control calibration system for a work machine including a frame, a linkage assembly and a work implement, the grade control calibration system, including: a vision processing system, including: an imaging device fixed to the linkage assembly and configured to generate a three-dimensional point cloud of a cutting portion of the work implement, and a vision controller in electronic communication with the imaging device and configured to calculate a current length of the cutting portion of the work implement based on the three-dimensional point cloud generated by the imaging device, the vision controller further configured to transmit the current length of the cutting portion of the work implement; and a grade control system in communication with the vision processing system and the linkage assembly, the grade control system including: a grading controller configured to receive the current length of the cutting portion of the work implement from the vision controller and to calibrate a position of the work implement relative to a terrain based on the received current length of the cutting portion of the work implement.
10. The grade control calibration system of claim 9, wherein the linkage assembly includes a boom, a stick and a plurality of electrohydraulic position sensors configured to transmit kinematics data to the vision controller.
11. The grade control calibration system of claim 10, wherein the vision controller determines whether the work implement is within a field of view of the imaging device using the kinematics data.
12. The grade control calibration system of claim 9, wherein the grade control system includes a memory module for storing a three-dimensional site map and a work tool profile, the work tool profile including an original length of the cutting portion of the work implement and a last measured length of the cutting portion of the work implement.
13. The grade control calibration system of claim 12, wherein the grading controller updates the last measured length of the cutting portion of the work implement to the current length of the cutting portion of the work implement in the memory module.
14. The grade control calibration system of claim 13, wherein the grading controller calibrates the grade control system using the site map and the updated last measured length of the cutting portion of the work implement.
15. A method of grading terrain using a work implement of a work machine, the work machine including a linkage assembly and a sensor coupled to the linkage assembly, the method comprising: detecting the work implement within a field of view of the sensor; identifying, by the sensor, a cutting portion of the work implement; calculating, by a controller electronically coupled to the sensor, a length of the cutting portion of the work implement; calibrating, by the controller, a grade control system of the work machine based on the calculated length of the cutting portion of the work implement; maneuvering, by the calibrated grade control system, the cutting portion of the work implement proximate the terrain; and grading the terrain with the cutting portion of the work

implement.

16. The method of claim 15, wherein the work implement is a bucket, and the cutting portion of the work implement is a plurality of teeth.

17. The method of claim 16, wherein the calculating further includes calculating, by the controller, the length of each of the plurality of teeth.

18. The method of claim 15, wherein the identifying further includes: generating, by the sensor, a current three-dimensional point cloud scan of the cutting portion of the work implement.

19. The method of claim 18, wherein the calculating further includes: retrieving, by the controller, a previous three-dimensional point cloud scan of the cutting portion of the work implement, and generating, by the controller, a disparity map of the cutting portion of the work implement based on the previous three-dimensional point cloud scan and the current three-dimensional point cloud scan.

20. The method of claim 19, wherein the calculating the length of the cutting portion of the work implement is based on the generated disparity map.
